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Tightly Secure Lattice Identity-Based Signature in the Quantum Random Oracle Model

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Abstract. We present a quantumly secure identity-based signature scheme based on the standard short integer solution problem, featuring tight security reductions in the quantum and classic random oracle models. The scheme has short signatures. Each signature contains a single lattice vector plus a single bit. Compared to the existing tightly secure, short lattice identity-based signature schemes by Pan and Wagner (PQCrypto'21), our scheme has a shorter signature size (around 30% shorter), stronger unforgeability, relies on a weaker assumption, and has detailed proof in the quantum random oracle model.

Keywords: Identity-based signature, strong unforgeability, lattice, quantum random oracle model

1 Introduction

Designing cryptographic schemes often follows a reductionist approach. More specifically, a cryptographic scheme is constructed along with a security theorem, stating that in some pre-defined security model, a t-time adversary who breaks the cryptographic scheme with probability ϵ can be turned into a t'-time algorithm that solves some computational problem with probability $\epsilon' \geq \epsilon/\theta$ with $\theta \geq 1$ and $t \approx t'$. If the problem is believed hard, ϵ' would be very small (or negligible in the security parameter in the asymptotic sense), making ϵ small too. As a result, the cryptographic scheme is unlikely broken. The factor θ measures the *tightness* of the reductionist proof. We say a reduction is tight if θ is a small *constant* and a cryptographic scheme with a tight security reduction is tightly secure. Tight security reduction is an attractive feature, with which, the cryptographic scheme can be implemented with smaller parameters (e.g., key sizes) to reach the same security level ϵ, leading to better efficiency.

Many cryptographic schemes have tight reductions. Among them, only a few tightly secure cryptographic schemes rely on quantum-immune assumptions, e.g., computational assumptions from high-dimensional lattices [6,3,2,4,15,16]. Following the recent works of Pan and Wagner, [15,16], we focus on quantumly secure identity-based signature (IBS) schemes with short signatures and tight security reductions from (believed) quantumly intractable lattice problems. Here

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"short signature" means a signature contains a constant number of non-zero lattice vectors. Under the umbrella of identity-based cryptography [20], IBS eases the issue of authenticating public keys and may replace PKI systems in small domains.

An IBS scheme has a master public key Mpk and a master private key Msk. The scheme allows using an arbitrary string id, called identity, as a user's public key. Messages are signed by a private key $\mathsf{Sk}_{\mathsf{id}}$, extracted using Msk and the identity id. Verifying the signatures produced by $\mathsf{Sk}_{\mathsf{id}}$ is done publicly using Mpk, id. The standard security for IBS is existential unforgeability under chosen-identity and chosen-message attacks. This security notion is defined via a security game. In the security game, the adversary can make key extraction queries to request private keys for any identities of its choice. The adversary can also make signing queries to request signatures on identity/message pairs. Finally, the adversary breaks the IBS scheme if it comes up with a valid message/signature pair (m^*, σ^*) under a challenge identity id^* provided the adversary has not obtained $\mathsf{Sk}_{\mathsf{id}^*}$ and a valid signature on m^* under id^*. To get tight security, a reductionist proof is required with the following two probabilities are constants close to 1:

1. the probability of successfully replying *all* key extraction and signing queries.
2. the probability that the adversary's forgery can be exploited to solve the underlying computational problem.

While post-quantum IBS schemes can be obtained from different approaches (e.g., via hierarchical identity-based encryption schemes, identification schemes, and generically from normal digital signature schemes), the ones with short signatures and tight security are rare: the lattice-based IBS schemes by Pan and Wagner [15,16] are the only schemes proposed to date.

Pan and Wagner obtained their IBS scheme in two steps. First, a weakly secure lattice IBS scheme is built based on lattice trapdoor delegation techniques [1,5]. The IBS scheme is weakly secure in the sense that the key extraction queries and signing queries must be committed before seeing the master public key. Second, two generic transforms are devised that tightly convert the weakly secure IBS scheme into IBS schemes with standard security. The IBS schemes by Pan and Wagner [15,16] fall short in several aspects. Firstly, they lack detailed security proofs in the quantum random oracle model (QROM), which are desirable for post-quantum cryptographic schemes [3][1]. QROM security proofs model cryptographic hash functions as truly random functions called quantum-accessible random oracles. It is assumed that a quantum adversary can access these random oracles via superpositions and obtains the output quantum states.

Secondly, the generic transforms devised in [15] add extra overhead to signatures and may complicate security analysis. More specifically, the standard-model transform introduced in [15,16] adds two randomnesses of lattice-based

[1] The authors sketch a QROM proof. Combined with their standard model transform, the resulting IBS may be proved tightly secure in the QROM. However, a detailed QROM proof remains missing.

chameleon hashing [5] and two security-parameter-sized pre-images of cryptographic hashing, respectively, to each signature from the weakly secure IBS scheme. The added overhead, especially the lattice-based chameleon hashing randomnesses, is certainly not negligible (please refer to Section 1.4 for details.). Meanwhile, the random-oracle-model transform does require extra analysis in the QROM due to the extra random oracles. No QROM analysis is provided to the random-oracle-model transform in [16].

We note that Lee et al. [12] show that the generic certification approach gives tightly secure IBS schemes using a digital signature scheme that is tightly secure in the multi-user setting with adaptive corruption [7]. The only known such signature scheme, due to Pan and Wagner [17], needs the signature to contain a public key with size quadratic in the security parameter (see Table 8, [17]), resulting in IBS scheme no longer having short signatures.

In summary, the current state motivates constructing simple, more efficient, short post-quantum IBS schemes with tight security reductions in the QROM.

1.1 Our Contributions

We construct a quantumly secure IBS scheme with short signatures and a tight security reduction, improving the schemes from [15,16]. We give a detailed *tight* reductionist proof, which reduces the hardness of the standard short integer solution (SIS) problem to the strong unforgeability[2] of the IBS scheme in the quantum random oracle model (QROM). QROM security proofs model cryptographic hash functions as truly random functions called quantum-accessible random oracles. It is assumed that a quantum adversary can access these random oracles via superpositions and obtains the output quantum states.

Our IBS scheme is also provably and tightly secure in the classic random oracle model. The proof is similar to the QROM proof, and we give the proof sketch in Section 5. Like the IBS schemes from [15], our IBS scheme can be adapted to lattices over polynomial rings with the potential benefit of having shorter key sizes. This comes from the fact that the integer lattice tools that we use (e.g., lattice trapdoor delegation and preimage samplings) are all adaptable to lattices over polynomial rings [13].

1.2 Our Techniques

We give an overview of the tightly secure lattice-based IBS scheme by Pan and Wagner [15,16], referred to as the PW scheme. The PW scheme is constructed in two steps. A weakly and tightly secure IBS scheme is constructed first. In the security proof of the scheme, the adversary has to announce its key extraction queries and signing queries before seeing the public key. The master public key of the scheme contains a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and two hash functions $H_1 : \{0,1\}^* \rightarrow$

[2] Strong unforgeability stipulates that even with a signature of the message, the adversary still cannot derive a new signature on that message. Unforgeability prevents the adversary from producing a new message and its valid signature.

$\mathbb{Z}_q^{n\times m}$ and $H_2 : \{0,1\}^* \to \mathbb{Z}_q^{n\times m}$ modelled as random oracles. The master private key is a trapdoor matrix $\mathbf{R}$ that efficiently solves the SIS problem defined by $\mathbf{A}$. A private key for identity id is a short lattice basis $\mathbf{T}_{\mathsf{id}}$ for the matrix $[\mathbf{A}|H(\mathsf{id})]$, obtained by the master secret key $\mathbf{R}$ via lattice trapdoor delegation [5,1,13]. To sign a message m with identity id, a short Gaussian vector $\mathbf{d}$ is computed using $\mathbf{T}_{\mathsf{id}}$ such that $[\mathbf{A}|H_1(\mathsf{id})|H_2(\mathsf{id},\mathsf{m})]\mathbf{d} = \mathbf{0}$. In the security proof, the random oracles H_1 and H_2 are programmed using the adversary's committed queries, ensuring all queries can be answered without failure, making the proof tight. Second, two generic transforms are designed to tightly lift the weakly secure scheme into fully secure IBS schemes.

Our scheme achieves tight security very differently. We use the Katz-Wang random-bit technique [10] and the properties of preimage-samplable functions (PSF) [6]. The master public key contains a matrix $\mathbf{A}$ and two hash functions $H_1 : \{0,1\}^* \to \mathbb{Z}_q^{n\times m}$ and $H_2 : \{0,1\}^* \to \mathbb{Z}_q^n$, which are modelled as (quantum) random oracles. Given an identity id, a random bit $b \in \{0,1\}$ is selected. The identity key $\mathsf{Sk}_{\mathsf{id}} := (\mathbf{R}_{\mathsf{id}||b}, b)$ where $\mathbf{R}_{\mathsf{id}||b}$ is a lattice trapdoor for the matrix $[\mathbf{A}|H_1(\mathsf{id}||b)]$ and '||' denotes string concatenation. A signature on the identity id and message m is $\sigma := (\mathbf{d}, b)$ where $\mathbf{d}$ is a short Gaussian vector $\mathbf{d}$ such that $[\mathbf{A}|H_1(\mathsf{id}||b)]\mathbf{d} = H_2(\mathsf{id}||\mathsf{m})$, and b is the bit value attached to the private key $\mathsf{Sk}_{\mathsf{id}}$. Here $\mathbf{d}$ is produced via preimage sampling [6,1,13]. We stress that in the real scheme, both keys $\mathsf{Sk}_{\mathsf{id}} := (\mathbf{R}_{\mathsf{id}||b}, b)$ and $\mathsf{Sk}_{\mathsf{id}} := (\mathbf{R}_{\mathsf{id}||1-b}, 1-b)$ can be constructed using Msk, but only one of them is given.

Making the change on $H_2(\mathsf{id}||\mathsf{m})$ from the PW scheme for our scheme has positive effects. First, a signature of our scheme contains a vector of a lattice with a smaller dimension (smaller by $n \log q$). This makes our signature size 25% to 30% shorter than those from [15]. Second, a shorter signature size allows a weaker SIS assumption (since the security proof can extract better SIS problem solutions). Third, it enables stronger unforgeability. We compare our scheme with the IBS schemes in [15,16] in Section 1.4.

Our scheme is not subject to the technical difficulty of using the Katz-Wang random-bit technique for tightly secure IBS mentioned by Pan and Wagner [15]. This is because such a technical difficulty stems from using the random-bit technique to partition both the identity space *and* the message space. In contrast, our scheme only partitions the identity space.

1.3 Overview of Our Security Proof

We sketch the security reductionist proof in the ROM to show our ideas. For simplicity, we assume that identical adversarial queries will receive the same reply in the security reduction. For example, making a key extraction query on the identity id twice will receive the same identity key. In our full scheme, we use the standard technique, i.e., (hash-based) pseudo-random functions, to maintain such consistency without making the scheme stateful.

The reduction starts with a SIS problem instance defined by a random matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and uses $\mathbf{A}$ to simulate the public parameters to the hypothetical IBS adversary. For each identity id, the reduction selects a random bit $\mu \in \{0,1\}$.

The random oracle query $\mathsf{id}||b$ to H_1 is answered by programming $H_1(\mathsf{id}||b)$ in a way that $H_1(\mathsf{id}||b)$ is uniformly random, and $[\mathbf{A}|H(\mathsf{id}||b)]$ has a lattice trapdoor if $b = \mu$ but has no lattice trapdoors if $b = 1 - \mu$.

Given a key extraction query on the identity id, the reduction programs the random oracle output $H_1(\mathsf{id}||\mu)$ and uses the trapdoor to get the identity key $\mathsf{Sk}_{\mathsf{id}||\mu}$. It is crucial that μ looks random to the adversary given $\mathsf{Sk}_{\mathsf{id}||\mu}$. Note, the reduction cannot produce the other key $\mathsf{Sk}_{\mathsf{id}||1-\mu}$. Only one of the two keys is given in the real scheme. To answer a signing query on message m identity id, the reduction returns a stored signature if it already exists. Otherwise, the reduction programs $H_2(\mathsf{id}||\mathsf{m})$ as follows.

1. Select a random bit b and get $H_1(\mathsf{id}||b)$ by calling the random oracle H_1.
2. Sample a Gaussian vector $\mathbf{d}$, program $H_2(\mathsf{id}||\mathsf{m})$ as $[\mathbf{A}|H_1(\mathsf{id}||b)]\mathbf{d}$, and return $(\mathbf{d}, b)$ as the signature.

We do the same to answer the random oracle queries to H_2. Given a query $\mathsf{id}||\mathsf{m}$ to H_2, the above steps are performed to program $H_2(\mathsf{id}||\mathsf{m})$ and then store the signature for future queries. Applying the regularity argument from [6], $H_2(\mathsf{id}||\mathsf{m})$ is distributed statistically close to the uniform distribution over $\mathbb{Z}_q^n$. The signature is distributed as if it was generated correctly using one of the (unknown) identity keys, either $\mathsf{Sk}_{\mathsf{id}} = (\mathbf{R}_{\mathsf{id}||\mu}, \mu)$ or $\mathsf{Sk}_{\mathsf{id}} = (\mathbf{R}_{\mathsf{id}||1-\mu}, 1 - \mu)$, as required. Thus, security reduction can always answer signing queries.

Finally, the security reduction exploits the adversary's forged signature, e.g., $\sigma^* = (\mathbf{d}^*, \delta^*)$, on the identity id^* and message m^* where $\mathbf{d}^*$ is of low-norm and $[\mathbf{A}|H_1(\mathsf{id}||\delta^*)]\mathbf{d}^* = H_2(\mathsf{id}^*||\mathsf{m}^*)$. We assume a query $H_2(\mathsf{id}^*||\mathsf{m}^*)$ is made before the forgery is produced (without querying $H_2(\mathsf{id}^*||\mathsf{m}^*)$, σ^* will be valid with negligible probability). Recall that id^* is associated with a random value μ^*, the reduction can produce the identity key for id^* and δ^* when $\delta^* = \mu^*$, and can use the forgery when $\delta^* = 1 - \mu^*$. To exploit such a forgery, we need:

1. The internal states used to program $H_2(\mathsf{id}^*||\mathsf{m}^*)$, i.e., the value b^*, $\mathbf{d}$ such that $[\mathbf{A}|H_1(\mathsf{id}^*||b^*)]\mathbf{d} \to H_2(\mathsf{id}^*||\mathsf{m}^*)$.
2. The internal states used to simulate $H_1(\mathsf{id}||b^*)$.
3. Crucially, μ^* and δ^* that satisfy $b^* = 1 - \mu^*$ and $\delta^* = 1 - \mu^*$.

We argue that identity-associated bit μ^* remains from the adversary's view. First, the adversary cannot make a key extraction query on id^*. So, it cannot obtain μ^* from $\mathsf{Sk}_{\mathsf{id}^*}$. Moreover, μ^* is not leaked via the signing queries since $(\mathsf{id}^*, \mathsf{m}^*)$ is not allowed to ask for a signature. And, according to the reduction, the signing query on $(\mathsf{id}^*, \mathsf{m} \neq \mathsf{m}^*)$ only gets a random b^* from the signature, which is independent of μ^*. Meanwhile, μ^* is not used anywhere else in the reduction. Thus, the conditions $b^* = 1 - \mu^*$ and $\delta^* = 1 - \mu^*$ are met with a probability of $1/4$. So, the reduction solves the SIS problem with a probability close to $\epsilon/4$, making the reduction tight.

To prove the security in the QROM, we follow the works by Boneh et al. [3] and Katsumata et al. [9]. We carefully maintain the internal states to make the proof historic-free, meaning that the random oracle query responses are made without looking up previously generated values and states.

1.4 Comparison

We compare our scheme with the only known tightly secure and short lattice IBS schemes by Pan and Wagner [15,16] in Table 1. The IBS schemes from [15] are constructed by applying the two generic transforms to the weakly secure PW IBS scheme. The first transform does not require random oracles. It uses the lattice chameleon hash function [5]. The second one is in the ROM. We denote the IBS schemes obtained via the first and second transforms by PW'21-I and PW'21-II, respectively. The comparison shows that our IBS scheme outperforms essentially in all aspects, including relying on a weaker SIS assumption, enjoying shorter key and signature sizes, and achieving stronger unforgeability.

Table 1: Comparison among tightly secure and short IBS schemes from latices

Schemes	SIS param. β	$\lvert\mathsf{Mpk}\rvert$	$\lvert\text{Signature}\rvert$	Strongly Unforgeable	Proof in QROM
PW'21-I [16]	$\tilde{O}(n^{2.5})$	$(2nm+2\lambda)\times V_{\mathbb{Z}_q}$	$3\times V_{\mathbb{Z}^m}+2\times V_{\mathbb{Z}^w}$	NO	YES
PW'21-II [15]	$\tilde{O}(n^{2.5})$	$nm\times V_{\mathbb{Z}_q}$	$2\times V_{\mathbb{Z}^{m+w}}+4\lambda$	NO	NO
This work	$\tilde{O}(n^{1.5})$	$nm\times V_{\mathbb{Z}_q}$	$1\times V_{\mathbb{Z}^{m+w}}+1$ bit	YES	YES

Let λ be the security parameter. The parameters n, m, q, $w := n\lceil\log q\rceil$ are standard in the context of lattice-based cryptography [13]. The three schemes in Table 1 are set with the same n, m, q. The parameter β represents the strength of the underlying SIS problem. The larger β becomes, the easier the SIS problem becomes; hence, the less secure the scheme becomes.[3] The $\tilde{O}$ notation hides the logarithmic factors. Table 1 shows that our scheme uses a more dificult SIS problem, which is equivalent to saying that our scheme relies on a weaker SIS assumption.

We remark that the sizes of the master private key Msk and the identity key $\mathsf{Sk}_{\mathsf{id}}$ of the schemes are respectively dominated by the size of the lattice trapdoor matrices in $\mathbb{Z}^{m\times w}$ and $\in \mathbb{Z}^{(m+w)\times w}$. In addition, our scheme's master private and private identity keys, respectively contain two and one double-security-parameter-sized random binary strings. Since we are working on the (quantum) random oracle model, these random strings can be efficiently and deterministically generated by hash-based pseudorandom functions (see Lemma 2) using Msk and $\mathsf{Sk}_{\mathsf{id}}$ and barely affect the tightness of the reductions. Therefore, the schemes have essentially the same sizes in master private and identity keys.

The size of the master public key and signature are denoted by $\lvert\mathsf{Mpk}\rvert$ and $\lvert\text{Signature}\rvert$, respectively. $V_{\mathbb{Z}_q}$ and $V_{\mathbb{Z}^\ell}$ represent the size of a $\mathbb{Z}_q$-vector and a $\mathbb{Z}^\ell$-vector, respectively.[4] The quantity $2\lambda\times V_{\mathbb{Z}_q}$ appeared under $\lvert\mathsf{Mpk}\rvert$ is from

[3] The parameter β for PW'21-I and PW'21-II are from Section 4, [15].

[4] For the IBS schemes, the effect of the Gaussian variances of signatures on signature size is rather limited as the distributions are centred at zero.

the lattice-based chameleon hash function [5] used by the transform for PW'21-I. For a message $\mathbf{m} \in \{0,1\}^\ell$ and a randomness $\mathbf{r} \in \mathbb{Z}^m$, The chameleon hash function is defined as $CHF(\mathbf{m};\mathbf{r}) := \mathbf{Br} + \mathbf{Cm}$ where $\mathbf{B} \in \mathbb{Z}_q^{n\times m}$ and $\mathbf{C} \in \mathbb{Z}_q^{n\times \ell}$. The chameleon hash function takes at least 2λ-bit inputs for λ-bit security due to the quantum algorithm Grover's search [8]. So, we set $\ell = 2\lambda$. The signature of PW'21-II has two random strings that are input to the random oracles. To ensure λ-bit security, the strings are at least 2λ bits long due to Grover's search.

PW'21-I and PW'21-II are not strongly unforgeable as multiplying a signature vector by a small constant, say 2, will likely result in a valid signature vector. Finally, after the publication of [15], [16] was updated with a sketched security proof of PW'21-I in the QROM. No QROM proof is given to PW'21-II.

2 Preliminaries

We denote the security parameter by λ. $\mathsf{poly}(\lambda)$ represents any polynomial in λ. We denote by $\mathsf{negl}(n)$ any negligible function in n. We use the standard asymptotic notations $O(\cdot)$, $\omega(\cdot)$, and $\Omega(\cdot)$. We use bold lowercase letters (e.g. $\mathbf{a}$) to denote vectors and bold capital letters (e.g. $\mathbf{A}$) to denote matrices. For an integer $q \geq 2$, let $\mathbb{Z}_q$ be the ring of integers modulo q. We denote by $[\mathbf{A}|\mathbf{B}]$ the concatenation of $\mathbf{A}$ and $\mathbf{B}$. We denote by $\|\mathbf{x}\|$ the ℓ_2 norm of a vector $\mathbf{x}$. $s_1(\mathbf{X})$ denotes the spectrum norm of the matrix $\mathbf{R}$, i.e., $s_1(\mathbf{R}) = \max_{\|\mathbf{u}\|=1} \|\mathbf{Xu}\|$.

If X is some distribution, we denote by $x \leftarrow X$ a sample x drawn according to the distribution X. We denote by $x \leftarrow U(\mathcal{X})$ drawing a sample x uniformly at random from a finite set $\mathcal{X}$. Let X and Y be two random variables taking values in some finite set S. We say random variables (or distributions) are statistically close or $\mathsf{negl}(\lambda)$-close if their statistical distance, $\Delta(X,Y) := \frac{1}{2}\sum_{s\in S} |\Pr[X = s] - \Pr[Y = s]|$ is negligible in the security parameter λ.

2.1 Quantum Computation

We follow a series of existing works [3,9,19,11] about cryptography in QROM to give a brief overview of quantum computation. A n qubits state $|\psi\rangle$ is expressed as $\sum_{x\in\{0,1\}^n} \alpha_x|x\rangle \in \mathbb{C}^{2^n}$ where $\{\alpha_x\}_{x\in\{0,1\}^n}$ satisfies $\sum_{x\in\{0,1\}^n} |\alpha_x|^2 = 1$ is a set of complex numbers. $\{|x\rangle\}_{x\in\{0,1\}^n}$ is called an orthonormal computational basis. An n-qubit state $|\psi\rangle$ can be measured using the computational basis, The measurement results in a classic bit string $x \in \{0,1\}^n$ with probability $|\alpha_x|^2$ and after the measurement, the state becomes $|x\rangle$. An evolution of quantum state can be described by a unitary matrix $\mathbf{U}$, which transforms $|x\rangle$ to $U|x\rangle$. A quantum algorithm is composed of quantum evolutions described by unitary matrices and measurements. Given any classicly computatble function f, there exists an unitary matrix $\mathbf{U}_f$ such that $\mathbf{U}_f|x,y\rangle = |x, f(x) \oplus y\rangle$. The running time of a quantum algorithm $\mathcal{A}$ is defined as the number of universal gates and measurements required for running $\mathcal{A}$.

The quantum random oracle model (QROM) is an idealised model in which a cryptographic hash function is modelled as a truly random function that can be accessed publicly and quantumly. In the security proof in the QROM, a random function $H : X \to Y$ is selected uniformly at random at the beginning of the security experiment (or security games). Every entity involved in the experiment is given access to an oracle, which given a state $\sum_{x,y} \alpha_{x,y}|x, y\rangle$ returns $\sum_{x,y} \alpha_{x,y}|x, H(x) \oplus y\rangle$. In the QROM, one query to the oracle is counted as one unit of time. As mentioned in [9], one random function H can be used to implement n random functions for n quantum (and classical) random oracles: for integer $0 \leq i \leq n-1$, the i-th random function is defined as $H_i(x) := H(i||x)$. We need the following two lemmas due to Boneh et al. [3] and Saito et al. [19].

Lemma 1. *Let $\epsilon > 0$. Say $\mathcal{A}$ is a quantum algorithm that makes Q quantum oracle queries. Suppose that we draw the oracle O from two distributions. The first is the random oracle distribution. The second is the distribution of oracles where the value of the oracle at each input x is identically and independently distributed by some distribution D whose statistical distance is within ϵ from uniform. Then the statistical distance between the distributions of outputs of $\mathcal{A}$ with each oracle is at most $4Q^2\sqrt{\epsilon}$.*

Lemma 2. *Let λ be the security parameter. Let ℓ be a positive integer. Let $H : \{0,1\}^\ell \times X \to Y$ and $H' : X \to Y$ be two independent random oracles. If an unbounded time adversary $\mathcal{A}$ who is given quantum access to H and H' makes at most Q_H queries to to H, then for $k \leftarrow U(\{0,1\})^\ell$ we have*

$$\left|\Pr[\mathcal{A}^{|H\rangle,|H(k,\cdot)\rangle}(1^\lambda) = 1] - \Pr[\mathcal{A}^{|H\rangle,|H'\rangle}(1^\lambda) = 1]\right| \leq 2Q_H \cdot 2^{-\ell/2}$$

According to Saito et al., [19], the above lemma also applies to the ROM, i.e., H and H' are classic random oracles.

2.2 Lattices and Discrete Gaussians

Definition 1. *For a positive integer q (later to be prime), a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, define*

$$\Lambda_q^\perp(\mathbf{A}) := \{\mathbf{y} \in \mathbb{Z}^m \ s.t. \ \mathbf{A}\mathbf{y} = \mathbf{0} \pmod q\}$$

as the m-dimensional full-rank integer lattices and

$$\Lambda_q^{\mathbf{u}}(\mathbf{A}) := \{\mathbf{e} \in \mathbb{Z}^m \ : \mathbf{A}\mathbf{e} = \mathbf{u} \pmod q\}$$

for $\mathbf{u} \in \mathbb{Z}_q^n$ as the shift of $\Lambda_q^\perp(\mathbf{A})$.

Definition 2. *Let $m \in \mathbb{Z}$ be a positive integer and $\Lambda \subseteq \mathbb{Z}^m$. For any real vector $\mathbf{c} \in \mathbb{R}^m$ and positive parameter $\sigma \in \mathbb{R}$, $\forall \mathbf{y} \in \Lambda$, the discrete Gaussian distribution over Λ with centre $\mathbf{0}$ and parameter σ is defined by $D_{\Lambda,\sigma} = \frac{\rho_\sigma(\mathbf{y})}{\rho_\sigma(\Lambda)}$ where $\rho_\sigma(\mathbf{x}) = \exp\left(-\pi\|\mathbf{x}\|^2/\sigma^2\right)$ is the Gaussian function and $\rho_\sigma(\Lambda) = \sum_{\mathbf{x}\in\Lambda} \rho_\sigma(\mathbf{x})$.*

Lemma 3 (Special case of Lemma 2.9, [13]). *Let n, m be positive integers, and $\mathbf{R} \leftarrow D_{\mathbb{Z},s}^{n\times m}$. There is a constant $C \approx 1/\sqrt{2\pi}$ such that for any $t \geq 0$, $s_1(\mathbf{R}) \leq C \cdot s \cdot (\sqrt{n} + \sqrt{m} + t)$ except with probability at most $2\exp(-\pi t^2)$.*

Lemma 4 (Lemma 4.4 of [14]). *For any lattice $\Lambda \subseteq \mathbb{Z}^n$, and reals $0 < \epsilon < 1$, $r \geq \omega(\sqrt{\log n})$, we have $\Pr_{\mathbf{x}\sim D_{\Lambda,r,\mathbf{c}}}[\|\mathbf{x}\| > r\sqrt{n}] \leq \frac{1+\epsilon}{1-\epsilon} \cdot 2^{-n}$.*

Lemma 5. *For any n-dimensional lattice $\Lambda \subseteq \mathbb{Z}^n$, positive $\epsilon < 1/3$, and $s \geq \omega(\sqrt{\log n})$, and for every $\mathbf{x} \in \Lambda$, the min-entropy $\mathrm{H}_\infty(D_{\Lambda,s}) \geq n - 1$.*

2.3 Lattice Trapdoors

Let $k = \lceil \log q \rceil$, $\mathbf{g}^\intercal = (1, 2, 4, \ldots, 2^{k-1})$. Define the n-by-nk gadget matrix as

$$\mathbf{G} = [\mathbf{g}^\intercal \otimes \mathbf{I}_n] = \begin{bmatrix} \mathbf{g}^\intercal & & & \\ & \mathbf{g}^\intercal & & \\ & & \ddots & \\ & & & \mathbf{g}^\intercal \end{bmatrix} \in \mathbb{Z}_q^{n\times nk}.$$

We recall the several lemmas that relates to the lattice trapdoor techniques. The lemma is a direct implication of Theorem 4.1.

Lemma 6. *There is a p.p.t algorithm $\mathsf{SampleR}(m, k, s)$ that takes as input $s \geq (\sqrt{\log n})$, and samples from a distribution statistically close to $D_{\mathbb{Z}^m,s}^k$.*

Let n, m, q, k be positive integers, where $m \geq 2n\log q + \omega(\sqrt{\log n})$. Let $w = \lceil n \log q \rceil$, $r = \omega(\sqrt{\log n})$. The first lemmas below are from Algorithm 1, Algorithm 3, and Algorithm 4 of [13]. The last lemma is from Proposition 5.1, [6].

Lemma 7. *There is a p.p.t algorithm $\mathsf{TrapGen}(1^n, 1^m, q)$ returns a matrix $\mathbf{A} = [\bar{\mathbf{A}}|\bar{\mathbf{A}}\mathbf{R} + \mathbf{H}\mathbf{G}] \in \mathbb{Z}_q^{n\times m}$ for $\bar{\mathbf{A}} \in \mathbb{Z}_q^{n\times(m-w)}$ and invertible $\mathbf{H} \in \mathbb{Z}_q^{n\times n}$ such that $\mathbf{A}$'s distribution is statistically close to $U(\mathbb{Z}_q^{n\times m})$ and the matrix $\mathbf{R}$'s distribution is statistically close to $D_{\mathbb{Z},r}^{(m-w)\times w}$. $\mathbf{R}$ is called a $\mathbf{G}$-trapdoor of $\mathbf{A}$.*

Lemma 8. *There is a p.p.t algorithm $\mathsf{SampleD}(\mathbf{A}, \mathbf{R}, \mathbf{U}, s)$ takes as input a $\mathbb{Z}_q^{n\times m}$-matrix $\mathbf{A}$ along with its $\mathbf{G}$-trapdoor $\mathbf{R}$, a vector $\mathbf{U} \in \mathbb{Z}_q^{n\times k}$ and $s \geq \omega(\sqrt{\log n}) \cdot s_1(\mathbf{R})$, and returns a vector $\mathbf{d} \in \mathbb{Z}^m$ which has a distribution statistically close to $D_{\mathbb{Z}^m,r}^k$ conditioned on $\mathbf{A}\mathbf{d} = \mathbf{U} \pmod q$, i.e., $D_{\Lambda_q^{\mathbf{U}}(\mathbf{A}),s}$.*

Lemma 9. *There is a p.p.t algorithm $\mathsf{DelTrap}(\mathbf{A}, \mathbf{R}, \mathbf{A}', \mathbf{H}, s')$ takes as input a $\mathbb{Z}_q^{n\times m}$-matrix $\mathbf{A}$, a $\mathbf{G}$-trapdoor $\mathbf{R}$ of $\mathbf{A}$, a $\mathbb{Z}_q^{n\times w}$-matrix $\mathbf{A}'$, and a Gaussian parameter s' where $s' \geq \omega(\sqrt{\log n}) \cdot s_1(\mathbf{R})$, and outputs a matrix $\mathbf{R}'$ whose distribution is statistically close to $D_{\mathbb{Z}^m,s'}^w$ conditioned on $\mathbf{A}' = \mathbf{A}\mathbf{R}' + \mathbf{H}\mathbf{G}$.*

Lemma 10. *For $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n\times m})$ and $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,s}$ with any $s \geq \omega(\sqrt{\log n})$, the distribution of $\mathbf{A}\mathbf{e} \bmod q$ is statistically close to $U(\mathbb{Z}_q^n)$. Furthermore, for a fixed $\mathbf{u} \in \mathbb{Z}_q^n$, the distribution of $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,s}$ given $\mathbf{A}\mathbf{e} = \mathbf{u} \bmod q$ is $D_{\Lambda_q^{\mathbf{u}}(\mathbf{A}),s}$.*

Our construction uses a special cases of Lemma 7 and Lemma 9 by setting the matrix $\mathbf{H} \in \mathbb{Z}_q^{n \times n}$ as the identity matrix, i.e., $\mathbf{I}_n$. Throughout this paper, we make the randomness used by the above algorithms explicit when needed and separate them from the primary inputs by semicolons. For example, $\mathsf{SampleR}(m, k, s; \phi)$, $\mathsf{SampleD}(\mathbf{A}, \mathbf{R}, \mathbf{U}, s; \varphi)$ and $\mathsf{DelTrap}(\mathbf{A}, \mathbf{R}, \mathbf{A}', \mathbf{H}, s; \psi)$, where ϕ, φ, and ψ denote the randomness used by these algorithms.

2.4 Short Integer Solution (SIS) Problem

Definition 3. *Let λ be the security parameter. The advantage of an algorithm $\mathcal{A}$ that solves the $\mathsf{SIS}_{n,q,m,\beta}$ problem, defined as,*

$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{SIS}_{n,q,m,\beta}}(\lambda) := \Pr\left[\mathcal{A}(\mathbf{A}, \beta) \rightarrow \mathbf{e} \neq \mathbf{0} \ : \ \mathbf{A}\mathbf{e} = 0 \bmod q \ \wedge \ \|\mathbf{e}\| \leq \beta\right]$$

where $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, We say (t, ϵ)-$\mathsf{SIS}_{n,q,m,\beta}$ assumption holds if for all t-time algorithms $\mathcal{A}$, $\mathsf{Adv}_{\mathcal{A}}^{\mathsf{SIS}_{n,q,m,\beta}}(\lambda) \leq \epsilon$.

Our IBS scheme uses the $\mathsf{SIS}_{n,q,m,\beta}$ problem with prime q and $\beta = \tilde{O}(n^{3/2})$, which is far from being reached by the known classic and quantum algorithms. We refer to Section 2.1, [18] for detailed discussions.

2.5 Identity-Based Signatures

Let λ be the security parameter. An identity-based signature (IBS) scheme $\mathcal{IBS}$ consists of four probabilistic polynomial time (p.p.t) algorithms.

- $\mathsf{Setup}(1^\lambda)$ returns a master public key Mpk and a master private key Msk.
- $\mathsf{Ext}(\mathsf{Mpk}, \mathsf{Msk}, \mathsf{id})$ generates an identity private key $\mathsf{Sk}_{\mathsf{id}}$.
- $\mathsf{Sig}(\mathsf{Mpk}, \mathsf{Sk}_{\mathsf{id}}, \mathsf{id}, \mathsf{m})$ returns a signature σ on a message m and identity id.
- $\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}, \mathsf{m}, \sigma)$ returns 1, indicating the signature σ is valid for the message m from a signer whose identity is id, or 0, otherwise.

We require for standard correctness, i.e., for all $(\mathsf{Mpk}, \mathsf{Msk}) \leftarrow \mathsf{Setup}(1^\lambda)$, $\mathsf{Sk}_{\mathsf{id}} \leftarrow \mathsf{Ext}(\mathsf{Mpk}, \mathsf{Msk}, \mathsf{id})$,

$$\Pr\left[\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}, \mathsf{m}, \mathsf{Sig}(\mathsf{Mpk}, \mathsf{Sk}_{\mathsf{id}}, \mathsf{id}, \mathsf{m}) = 1\right] \geq 1 - \mathsf{negl}(\lambda)$$

where $\mathsf{negl}(\lambda)$ is negligible in λ.

We recall the strong existential unforgeability under the chosen-identity attack and the chosen-message attack (SEUF-ID-CMA) for IBS. The notion is strong in the sense that no adversary can forge a signature on a message-identity pair from which the adversary has seen valid signatures. A weaker unforgeability notion (EUF-ID-CMA) as considered in [15] only requires that a forgery cannot be produced for a message-identity pair that have not been signed.

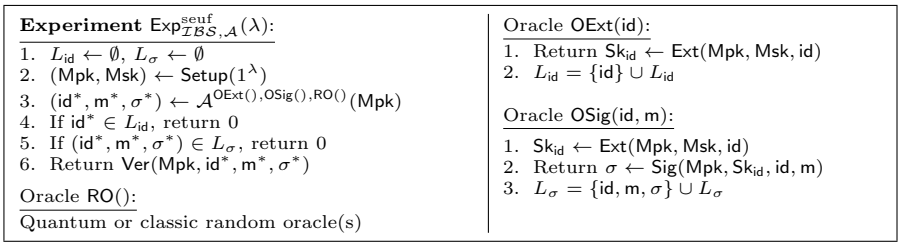


Fig. 1: SEUF-ID-CMA Security Experiment

Definition 4. *Consider the security experiment (game) defined in 1. We say that an IBS scheme $\mathcal{IBS}$ is $(t, Q_{\mathsf{Ext}}, Q_{\mathsf{Sig}}, \epsilon)$-SEUF-ID-CMA secure if for any t-time adversary $\mathcal{A}$ that makes at most Q_{Ext} (key extraction) queries to the oracle $\mathsf{OExt}()$ and at most Q_{Sig} (signing) queries to the oracle $\mathsf{OSig}()$, we have*

$$\mathsf{Adv}^{\mathsf{seuf}}_{\mathcal{IBS},\mathcal{A}}(\lambda) := \Pr[\mathsf{Exp}^{\mathsf{seuf}}_{\mathcal{IBS},\mathcal{A}}(\lambda) = 1] \leq \epsilon$$

where the probability is over the randomness of $\mathcal{A}$ and the IBS scheme.

3 Our IBS Scheme

We provide our IBS scheme Σ in Fig. 2. The scheme uses the following global parameters and cryptographic hash functions:

- Gadget matrix $\mathbf{G}$'s column dimension $w := n\lceil \log q \rceil$.
- Dimension $m \geq 2n \log q + \omega(\sqrt{\log n})$ for using Lemma 9.
- Gaussian parameter $r = \omega(\sqrt{\log n})$ for the algorithm $\mathsf{TrapGen}$.
- Gaussian parameter $s = O(\sqrt{w}) \cdot \omega(\sqrt{\log n})$ for algorithm $\mathsf{DelTrap}$.
- Gaussian parameter $s' = O(w) \cdot \omega(\sqrt{\log n})^2$ for algorithm $\mathsf{SampleD}$.
- Identity space $\{0,1\}^{\ell_{\mathsf{id}}}$ and message space $\{0,1\}^{\ell_{\mathsf{m}}}$ for $\ell_{\mathsf{id}}, \ell_{\mathsf{m}}$.
- ℓ_s and $\ell_{s'}$, the randomness bit lengths of $\mathsf{DelTrap}$ and $\mathsf{SampleD}$, respectively.
- Cryptographic hash functions modelled as (quantum) random oracles:
  - $\mathsf{H}_1 : \{0,1\}^\ell \times \{0,1\}^{\ell_{\mathsf{id}}} \to \{0,1\}$, $\mathsf{H}_2 : \{0,1\}^\ell \times \{0,1\}^{\ell_{\mathsf{id}}}\{0,1\}^{\ell_s}$, $\mathsf{H}_3 : \{0,1\}^\ell \times \{0,1\}^{\ell_{\mathsf{id}}+\ell_{\mathsf{m}}} \to \{0,1\}^{\ell_{s'}}$
  - $H_1 : \{0,1\}^{\ell_{\mathsf{id}}+1} \to \mathbb{Z}_q^{n \times w}$, $H_2 : \{0,1\}^{\ell_{\mathsf{id}}+\ell_{\mathsf{m}}} \to \mathbb{Z}_q^n$

In the scheme, each user (with identity id) also chooses and keeps a secret random string $k_{\mathsf{id}} \leftarrow U(\{0,1\}^\ell)$ for signing. $\mathsf{H}_1(k_1, \cdot)$, $\mathsf{H}_2(k_2, \cdot)$ and $\mathsf{H}_3(k_{\mathsf{id}}, \cdot)$ are used to make the signing and key extraction processes deterministic, i.e., if the same message and identity are submitted for signing (resp. key extraction), the same signature (resp. identity key) will be returned.

$\underline{\mathsf{Setup}(1^\lambda)}$:
1. $(\mathbf{A}, \mathbf{R}) \leftarrow \mathsf{TrapGen}(1^n, 1^m, q)$
2. $k_1, k_2 \leftarrow U(\{0,1\}^\ell)$
3. Return $\mathsf{Mpk} := \mathbf{A}$, $\mathsf{Msk} := \{\mathbf{R}, k_1, k_2\}$

$\underline{\mathsf{Sig}(\mathsf{Mpk}, \mathsf{Sk_{id}}, \mathsf{id}, \mathsf{m})}$:
*User id keeps a secret $k_{\mathsf{id}} \leftarrow U(\{0,1\}^\ell)$
1. Parse $\mathsf{Sk_{id}} = (\mathbf{R}_{\mathsf{id}||b}, b)$
2. $\mathbf{u} \leftarrow H_2(\mathsf{id}||\mathsf{m})$
3. $\mathbf{A}_{\mathsf{id}||b} \leftarrow H_1(\mathsf{id}||b)$, $\mathbf{F}_{\mathsf{id}||b} := [\mathbf{A}|\mathbf{A}_{\mathsf{id}||b}]$
4. $\varphi \leftarrow \mathsf{H}_3(k_{\mathsf{id}}, \mathsf{id}||\mathsf{m})$
5. $\mathbf{d} \leftarrow \mathsf{SampleD}(\mathbf{F}_{\mathsf{id}||b}, \mathbf{R}_{\mathsf{id}||b}, \mathbf{u}, s'; \varphi)$
6. Return $\sigma := (\sigma_1, \sigma_2) = (\mathbf{d}, b)$

$\underline{\mathsf{Ext}(\mathsf{Mpk}, \mathsf{Msk}, \mathsf{id})}$:
1. $b \leftarrow \mathsf{H}_1(k_1, \mathsf{id})$, $\psi \leftarrow \mathsf{H}_2(k_2, \mathsf{id})$
2. $\mathbf{A}_{\mathsf{id}||b} \leftarrow H_1(\mathsf{id}||b)$
3. $\mathbf{R}_{\mathsf{id}||b} \leftarrow \mathsf{DelTrap}(\mathbf{A}, \mathbf{R}, \mathbf{A}_{\mathsf{id}||b}, \mathbf{I}_n, s; \psi)$
4. Return $\mathsf{Sk_{id}} := (\mathbf{R}_{\mathsf{id}||b}, b)$

$\underline{\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}, \sigma, \mathsf{m})}$:
1. Parse $\sigma := (\mathbf{d}, b)$
2. $\mathbf{u} \leftarrow H_2(\mathsf{id}||\mathsf{m})$
3. Return 0 if $[\mathbf{A}|H_1(\mathsf{id}||b)]\mathbf{d} \neq \mathbf{u}$
4. Return 0 if $\|\mathbf{d}\| > s'\sqrt{m+w}$
5. Return 1

Fig. 2: The Proposed IBS Scheme

Correctness. We show that $\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}, \mathsf{m}, \mathsf{Sig}(\mathsf{Mpk}, \mathsf{Sk_{id}}, \mathsf{id}, \mathsf{m}) = 1$ happens with all but negligible probability. By Lemma 8, $\mathsf{Sig}(\mathsf{Mpk}, \mathsf{Sk_{id}}, \mathsf{id}, \mathsf{m})$ returns a signature $\sigma := (\mathbf{d}, b)$ where $\mathbf{d}$ distributes statistically close to $D_{\mathbb{Z},s'}^{m+w}$, conditioned on $[\mathbf{A}|H_1(\mathsf{id}||b)]\mathbf{d} = H_2(\mathsf{id}||\mathsf{m})$. So, step 3 of the algorithm $\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}, \sigma, \mathsf{m})$ returns 0 with negligible probability. Meanwhile, by Lemma 4, $\|\mathbf{d}\| \leq s'\sqrt{m+w}$ with all but negligible probability, which proves that $\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}, \sigma, \mathsf{m})$ returns 1, i.e., accepts the correctly generated signature, with all but negligible probability.

4 Security Proof in the Quantum Random Oracle Model

Theorem 1. *Let H_1, H_2, H_1, H_2, H_3 be random oracles that may be quantumly accessible. If (t, ϵ)-$\mathsf{SIS}_{n,q,m,\beta}$ assumption holds with $\beta = O(w^{3/2}) \cdot \omega(\sqrt{\log n})^2$, the IBS scheme Σ is $(t', Q_{\mathsf{Ext}}, Q_{\mathsf{Sig}}, 4\epsilon + \mathsf{negl}(\lambda))$-SEUF-ID-CMA secure for arbitrary Q_{Ext}, Q_{Sig}, and any $t' \leq t + O(Q_{H_1} + Q_{H_2}) \cdot \mathsf{poly}(\lambda)$ where Q_{H_1} and Q_{H_2} are the number of quantum random oracle queries to $|H_1\rangle$ and $|H_2\rangle$, respectively.*

We split our security proof into two steps. First, we devise a stateful version of our original IBS scheme Σ shown in Fig. 2, denoted by Σ', which preserves the security from Σ. Then, we reduce the hardness of $\mathsf{SIS}_{a,q,n,\beta}$ problem to the SEUF-ID-CMA security of the stateful IBS scheme Σ'.

4.1 Security Reduction from IBS Scheme Σ' to IBS Scheme Σ

The stateful scheme, as described in Fig. 3, removes H_1, H_2, H_3, and uses internal states to make consistent replies to repeated key extracting and signing queries. The (pseudo)random values produced by H_1, H_2, H_3 in the original scheme are instead sampled uniformly at random from the corresponding sets.

Considering the SEUF-ID-CMA security experiment (Fig. 1), it is clear that in both the original IBS scheme Σ and the stateful IBS scheme Σ', the same query id to OExt is responded by the same identity key, and the same query

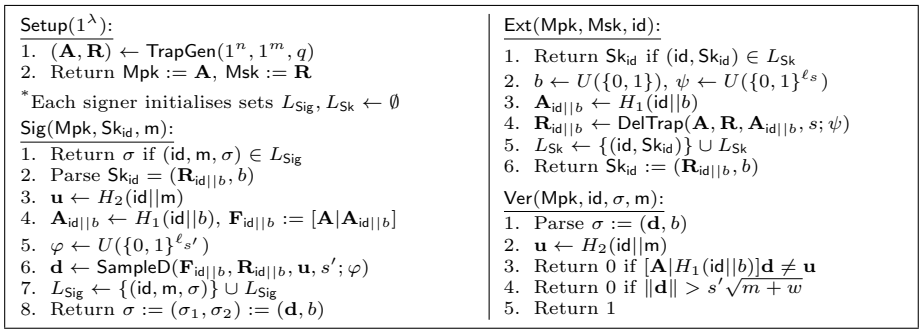


Fig. 3: The Stateful IBS Scheme Σ'

$(\mathsf{id}, \mathsf{m})$ to OSig is responded with the same signature value σ. Therefore, the views of the adversary $\mathcal{A}$ to Σ and Σ' are identical except how $b \in \{0,1\}$ and the randomness for $\mathsf{SampleD}$ and $\mathsf{DelTrap}$ are generated, i.e., generated using H_1, H_2, H_3 versus generated uniformly at random. We note that queries to OSig trigger queries to $\mathsf{H}_1(k_1, \cdot)$, and queries to OExt trigger queries to $\mathsf{H}_2(k_2, \cdot)$ and $\mathsf{H}_3(k_3, \cdot)$[5]. By Lemma 2 and union bound, a simple reduction shows

$$|\mathsf{Adv}^{\mathsf{seuf}}_{\Sigma,\mathcal{A}}(\lambda) - \mathsf{Adv}^{\mathsf{seuf}}_{\Sigma',\mathcal{A}}(\lambda)| \leq 2(2Q_{\mathsf{Ext}} + Q_{\mathsf{Sig}}) \cdot 2^{-\ell/2} \tag{1}$$

which is negligible in λ for polynomial ℓ (in λ), Q_{Ext} and Q_{Sig}. The reduction runs in time nearly the same as the run time of the adversary against Σ plus the time taken to simulate the parameters of Σ, which is polynomial in λ.

4.2 The Security of IBS Scheme Σ'

Proof. Assume there is a t-time SEUF-ID-CMA adversary who makes at most Q_{H_1} and Q_{H_2} queries, respectively to $\big|H_1\rangle$ and $\big|H_2\rangle$ and has advantage $\mathsf{Adv}^{\mathsf{seuf}}_{\Sigma',\mathcal{A}}(\lambda)$ in breaking the IBS Scheme Σ'. We use algorithm $\mathsf{SampleR}(k_1, k_2, s; \mathsf{coin})$ whose output distribution is statistically close to $D^{k_2}_{\mathbb{Z}^{k_1}, s}$. coin denotes the randomness: when $k_1 = m$, $k_2 = w$, $|\mathsf{coin}| = \ell_c$; when $k_1 = m + w$, $k_2 = 1$, $|\mathsf{coin}| = \ell_{c'}$.

We proceed with a sequence of security games defined in Fig. 4. Three procedures can describe each security game: Initialisation (defined in Fig. 5), Setup (defined in Fig. 6), and Ver. Each security game has four oracles: the key extraction oracle OExt (defined in Fig. 10), the signing oracle OSig (defined in Fig. 9); and two quantum random oracles $\big|H_1\rangle$, $\big|H_2\rangle$ (defined in Fig. 7 and Fig. 8, respectively). Ver is identical to the verification algorithm given in Fig. 2 and is kept the same across all games. The process Initialisation is included in the proof where random functions are set as random oracles accessed by the participants.

Let S_i be the event that the ith security game returns a well-defined value 1. The first security game G_0 runs the stateful IBS scheme Σ' according to the

[5] The adversary does not have direct oracle access to $\mathsf{H}_1(k_1, \cdot)$, $\mathsf{H}_2(k_2, \cdot)$ and $\mathsf{H}_3(k_3, \cdot)$.

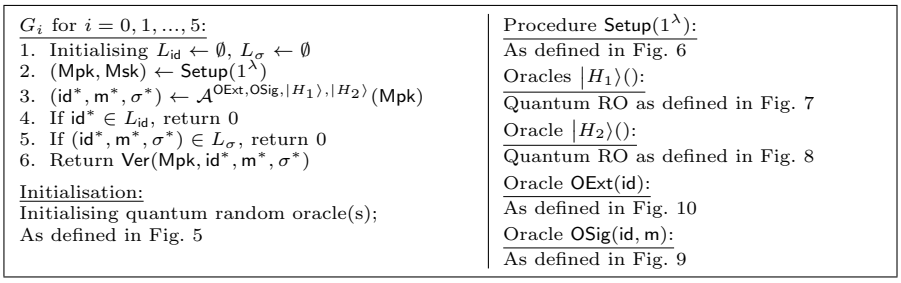


Fig. 4: Security games G_i for proofs in QROM

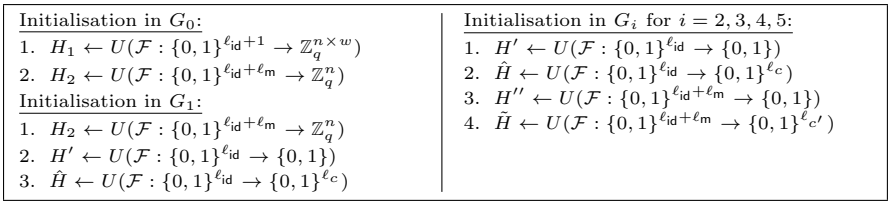


Fig. 5: Functions for implementing quantum random oracles in G_i

SEUF-ID-CMA security experiment (Fig. 1). In the QROM, cryptographic hash functions H_1 and H_2 are initialised as truly random functions and maintained by the reduction. By our hypothesis, we have

$$\mathsf{Adv}^{\mathsf{seuf}}_{\Sigma',\mathcal{A}}(\lambda) := \Pr[\mathsf{Exp}^{\mathsf{seuf}}_{\Sigma',\mathcal{A}}(\lambda) = 1] = \Pr[S_0] \tag{2}$$

Game G_1 changes how H_1 is simulated, as shown in Fig. 5 and Fig. 7. In particular, in G_1 a random function, $\hat{H}$ is selected to produce the randomness for the algorithm $\mathsf{SampleR}$. A random function H' is selected to help simulate H_1. According to the specification of H_1 for G_1 from Fig. 7, we have

$$H_1(\mathsf{id}||b) := \begin{cases} \mathbf{A}\mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id})||b) + \mathbf{G}, & \text{if } b = H'(\mathsf{id}) \\ \mathbf{A}\mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id})||b), & \text{otherwise} \end{cases}$$

Using Lemma 10 with the facts that matrix $\mathbf{A}$'s column dimension $m \geq 2n \log q + \omega(\sqrt{\log n})$ and parameter $s \geq \omega(\sqrt{\log n})$, $\mathbf{A}\mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id})||b)$, and thus, $H_1(\mathsf{id}||b)$ are distributed $\mathsf{negl}'_1(\lambda)$-close to $U(\mathbb{Z}_q^{n \times w})$. Since H_1 is queried Q_{H_1}

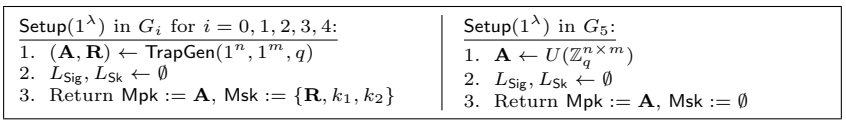


Fig. 6: Setup for the G_i

Oracle $|H_1\rangle(\cdot)$ in G_0,:

1. On input superposition $\sum_{\mathsf{id},b,y} \alpha_{\mathsf{id},b,y}|\mathsf{id}||b,y\rangle$, return

$$\sum\nolimits_{\mathsf{id},b,y} \alpha_{\mathsf{id},b,y}|\mathsf{id}||b, H_1(\mathsf{id}||b) \oplus y\rangle$$

Oracle $|H_1\rangle(\cdot)$ in G_i for $i = 1,2,3,4,5$:

Set $\mathbf{R}_{\mathsf{id}||b} \leftarrow \mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id}||b))$ and define H_1 as:

$$H_1(\mathsf{id}||b) := \begin{cases} \mathbf{A}\mathbf{R}_{\mathsf{id}||b} + \mathbf{G}, & \text{if } b = H'(\mathsf{id}) \\ \mathbf{A}\mathbf{R}_{\mathsf{id}||b}, & \text{otherwise} \end{cases}$$

1. On input superposition $\sum_{\mathsf{id},b,y}, \alpha_{\mathsf{id},b,y}|\mathsf{id}||b,y\rangle$, return

$$\sum\nolimits_{\mathsf{id},b,y} \alpha_{\mathsf{id},b,y}|\mathsf{id}||b, H_1(\mathsf{id}||b) \oplus y\rangle$$

Fig. 7: $|H_1\rangle$ for game G_i

Oracle $|H_2\rangle(\cdot)$ in G_i for $i = 0,1$:

1. On input superposition $\sum_{\mathsf{id},\mathsf{m},y} \alpha_{\mathsf{id},\mathsf{m},y}|\mathsf{id}||\mathsf{m},y\rangle$, return

$$\sum\nolimits_{\mathsf{id},\mathsf{m},y} \alpha_{\mathsf{id},\mathsf{m},y}|\mathsf{id}||\mathsf{m}, H_2(\mathsf{id}||\mathsf{m}) \oplus y\rangle$$

Oracle $|H_2\rangle(\cdot)$ in G_i for $i = 2,3,4,5$:

Set $\delta \leftarrow H''(\mathsf{id}||\mathsf{m})$ and define H_2 as:

$$H_2(\mathsf{id}||\mathsf{m}) := [\mathbf{A}|H_1(\mathsf{id}||\delta)]\mathsf{SampleR}(m + w, 1, s'; \tilde{H}(\mathsf{id}||\mathsf{m}))$$

1. On input superposition $\sum_{\mathsf{id},\mathsf{m},y} \alpha_{\mathsf{id},\mathsf{m},y}|\mathsf{id}||\mathsf{m},y\rangle$, return

$$\sum\nolimits_{\mathsf{id},\mathsf{m},y} \alpha_{\mathsf{id},\mathsf{m},y}|\mathsf{id}||\mathsf{m}, H_2(\mathsf{id}||\mathsf{m}) \oplus y\rangle$$

Fig. 8: $|H_2\rangle$ for game G_i

times. Using Lemma 1, we have for some negligible function $\mathsf{negl}_1(\lambda)$

$$|\Pr[S_1] - \Pr[S_0]| \leq 4Q_{H_1}^2 \sqrt{\mathsf{negl}_1'(\lambda)} \leq \mathsf{negl}_1(\lambda) \tag{3}$$

G_2 changes how H_2 is simulated. Two random functions are initialised, as shown in Fig. 5. $H'' : \{0,1\}^{\ell_{\mathsf{id}}+\ell_{\mathsf{m}}} \to \{0,1\}$) is used to specify the bit value that appears in the signature of $(\mathsf{id}, \mathsf{m})$. $\tilde{H} : \{0,1\}^{\ell_{\mathsf{id}}+\ell_{\mathsf{m}}} \to \{0,1\}^{\ell_{c'}}$ maps bit strings of length $\ell_{\mathsf{id}} + \ell_{\mathsf{m}}$ into $\ell_{c'}$-bit randomnesses for $\mathsf{SampleR}$. According to Fig. 8,

$$H_2(\mathsf{id}||\mathsf{m}) := [\mathbf{A}|H_1(\mathsf{id}||\delta)]\mathsf{SampleR}(m + w, 1, s'; \tilde{H}(\mathsf{id}||\mathsf{m}))$$

where $\delta \leftarrow H''(\mathsf{id}||\mathsf{m})$. Since H_1 is properly simulated (i.e., its output distribution is statistically close to $U(\mathbb{Z}_q^{n\times m})$), and $\mathsf{SampleR}(m+w, 1, s'; \tilde{H}(\mathsf{id}||\mathsf{m}))$ distributes statistically close to $D_{\mathbb{Z}^m,s'}$ where $s' \geq \omega(\sqrt{\log n})$, by Lemma 10, $H_2(\mathsf{id}||\mathsf{m})$ is

$\mathsf{OSig}(\mathsf{id},\mathsf{m})$ in G_i for $i=0,1,2$:	$\mathsf{OSig}(\mathsf{id},\mathsf{m})$ in G_i for $i=3,4,5$:
1. Return σ if $(\mathsf{id},\mathsf{m},\sigma) \in L_{\mathsf{Sig}}$ 2. $\mathsf{Sk}_{\mathsf{id}\|\|b} \leftarrow \mathsf{OExt}(\mathsf{id}) := \mathsf{Sk}_{\mathsf{id}} = (\mathbf{R}_{\mathsf{id}\|\|b}, b)$ 3. $\mathbf{u} \leftarrow H_2(\mathsf{id}\|\|\mathsf{m})$ 4. $\mathbf{A}_{\mathsf{id}\|\|b} \leftarrow H_1(\mathsf{id}\|\|b)$, $\mathbf{F}_{\mathsf{id}\|\|b} := [\mathbf{A}\|\mathbf{A}_{\mathsf{id}\|\|b}]$ 5. $\varphi \leftarrow U(\{0,1\}^{\ell_{s'}})$ 6. $\mathbf{d} \leftarrow \mathsf{SampleD}(\mathbf{F}_{\mathsf{id}\|\|b}, \mathbf{R}_{\mathsf{id}\|\|b}, \mathbf{u}, s'; \varphi)$ 7. $L_{\mathsf{Sig}} \leftarrow \{(\mathsf{id},\mathsf{m},\sigma)\} \cup L_{\mathsf{Sig}}$ 8. Return $\sigma := (\sigma_1,\sigma_2) := (\mathbf{d}, b)$ 9. $L_\sigma \leftarrow \{(\mathsf{id},\mathsf{m},\sigma)\} \cup L_\sigma$	1. Return σ if $(\mathsf{id},\mathsf{m},\sigma) \in L_{\mathsf{Sig}}$ 2. $\delta \leftarrow H''(\mathsf{id}\|\|\mathsf{m})$ 3. $\mathbf{d} \leftarrow \mathsf{SampleR}(m+w, 1, s; \tilde{H}(\mathsf{id}\|\|\mathsf{m}))$ 4. $L_{\mathsf{Sig}} \leftarrow \{(\mathsf{id},\mathsf{m},\sigma)\} \cup L_{\mathsf{Sig}}$ 5. Return $\sigma := (\sigma_1,\sigma_2) = (\mathbf{d}, \delta)$ 6. $L_\sigma \leftarrow \{(\mathsf{id},\mathsf{m},\sigma)\} \cup L_\sigma$

Fig. 9: Oracle $\mathsf{OSig}()$ for game G_i

$\mathsf{OExt}(\mathsf{id})$ in G_i for $i=0,1,2,3$:	$\mathsf{OExt}(\mathsf{id})$ in G_4, G_5:
1. Return $\mathsf{Sk}_{\mathsf{id}}$ if $(\mathsf{id},\mathsf{Sk}_{\mathsf{id}}) \in L_{\mathsf{Sk}}$ 2. $b \leftarrow U(\{0,1\})$, $\psi \leftarrow U(\{0,1\}^{\ell_s})$ 3. $\mathbf{A}_{\mathsf{id}\|\|b} \leftarrow H_1(\mathsf{id}\|\|b)$ 4. $\mathbf{R}_{\mathsf{id}\|\|b} \leftarrow \mathsf{DelTrap}(\mathbf{A}, \mathbf{R}, \mathbf{A}_{\mathsf{id}\|\|b}, s; \psi)$ 5. Return $\mathsf{Sk}_{\mathsf{id}} := (\mathbf{R}_{\mathsf{id}\|\|b}, b)$ 6. $L_{\mathsf{Sk}} \leftarrow \{(\mathsf{id},\mathsf{Sk}_{\mathsf{id}})\} \cup L_{\mathsf{Sk}}$ 7. $L_{\mathsf{id}} \leftarrow \{\mathsf{id}\} \cup L_{\mathsf{id}}$	1. Return $\mathsf{Sk}_{\mathsf{id}}$ if $(\mathsf{id},\mathsf{Sk}_{\mathsf{id}}) \in L_{\mathsf{Sk}}$ 2. $b \leftarrow H'(\mathsf{id})$ 3. $\mathbf{R}_{\mathsf{id}\|\|b} \leftarrow \mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id}\|\|b)$ 4. Return $\mathsf{Sk}_{\mathsf{id}} := (\mathbf{R}_{\mathsf{id}\|\|b}, b)$ 5. $L_{\mathsf{Sk}} \leftarrow \{(\mathsf{id},\mathsf{Sk}_{\mathsf{id}})\} \cup L_{\mathsf{Sk}}$ 6. $L_{\mathsf{id}} \leftarrow \{\mathsf{id}\} \cup L_{\mathsf{id}}$

Fig. 10: Oracle $\mathsf{OExt}()$ for game G_i

distributed $\mathsf{negl}'_2(\lambda)$-close to $U(\mathbb{Z}_q^n)$. Using Lemma 1, for some negligible $\mathsf{negl}_2(\lambda)$

$$|\Pr[S_2] - \Pr[S_1]| \leq 4Q_{H_2}^2 \sqrt{\mathsf{negl}'_2(\lambda)} \leq \mathsf{negl}_2(\lambda) \tag{4}$$

G_3 changes OSig, as specified in Fig. 9[6]. Assume the signature of (id,m) is $\sigma := (\mathbf{d}, b)$. In G_2, b is uniformly random. In G_3, $\delta \leftarrow H''(\mathsf{id}\|\|\mathsf{m})$ for random function H''. So, b is distributed identically to δ. In G_2, according to Lemma 8, the vector $\mathbf{d} \leftarrow \mathsf{SampleD}([\mathbf{A}|\mathbf{A}_{\mathsf{id}\|\|b}], \mathbf{R}_{\mathsf{id}\|\|b}, H_2(\mathsf{id}\|\|\mathsf{m}), s')$ has a distribution statistically close to $D_{\mathbb{Z}^{m+w},s'}$ conditioned on $[\mathbf{A}|H_1(\mathsf{id}\|\|b)]\mathbf{d} = H_2(\mathsf{id}\|\|\mathsf{m})$. In G_3, $\mathbf{d} \leftarrow \mathsf{SampleR}(m+w, 1, s; \tilde{H}(\mathsf{id}\|\|\mathsf{m}))$. By Lemma 10, $\mathbf{d}$ has a distribution statistically close to $D_{\mathbb{Z}^{m+w},s'}$. Also, $H_2(\mathsf{id}\|\|\mathsf{m}) := [\mathbf{A}|H_1(\mathsf{id}\|\|\delta)]\mathbf{d}$ where δ is uniformly random, as b. So, the distribution of $(\mathbf{d}, b)$ in G_2 and the distribution of $(\mathbf{d}, \delta)$ are negligibly close. So, for some negligible $\mathsf{negl}_3(\lambda)$

$$|\Pr[S_3] - \Pr[S_2]| \leq \mathsf{negl}_3(\lambda) \tag{5}$$

G_4 changes how OExt works, as specified in Fig. 10. Recall that for OExt in G_3, the bit value b is uniformly random. For OExt in G_4, $b \leftarrow H'(\mathsf{id})$. Since H' is truly random and not accessible to the adversary, the distributions of b in G_3, G_4 are identical. Moreover, in G_3, for fixed $\mathbf{A}_{\mathsf{id}\|\|b}$, $\mathbf{R}_{\mathsf{id}\|\|b}$ is sampled according to $\mathsf{DelTrap}(\mathbf{A}, \mathbf{R}, \mathbf{A}_{\mathsf{id}\|\|b}, s)$, i.e., $\mathbf{R}_{\mathsf{id}\|\|b}$ has a distribution that is statistically close to $D^w_{\mathbb{Z}^m,s}$, conditioned on $\mathbf{A}_{\mathsf{id}\|\|b} = \mathbf{A}\mathbf{R}_{\mathsf{id}\|\|b} + \mathbf{G}$. As per

[6] We note that in Fig. 9 the lists L_{Sig} and L_σ are identical, both storing signed messages, the corresponding identities and signatures. L_{Sig} is part of the signing algorithm (to make signing stateful), and the security games maintain L_σ.

Lemma 9, $\mathbf{R}_{\mathsf{id}||b}$ has a distribution statistically close to $D_{\Lambda_q^{\mathbf{A}_{\mathsf{id}||b}-\mathbf{G}}(\mathbf{A}),s}$. In G_4, $\mathbf{R}_{\mathsf{id}||b} \leftarrow \mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id}||b)$ which distributes statistically close to $D^w_{\mathbb{Z}^m,s}$. Meanwhile, according to the simulation of H_1, $H_1(\mathsf{id}||b) = H_1(\mathsf{id}||H'(\mathsf{id})) := \mathbf{A}_{\mathsf{id}||b} = \mathbf{A}\mathbf{R}_{\mathsf{id}||b} + \mathbf{G}$. As per Lemma 10, the distributions of $\mathbf{R}_{\mathsf{id}||b}$ in G_3 and G_4 are $\mathsf{negl}_4(\lambda)$-close for some $\mathsf{negl}_4(\lambda)$.

$$|\Pr[S_4] - \Pr[S_3]| \leq \mathsf{negl}_4(\lambda) \tag{6}$$

G_5 changes how Setup works, as showed by Fig. 6. In particular, the public matrix $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n\times m})$ and Msk is set empty. By Lemma 7, $\mathcal{A}$'s distributions in G_4 and G_5 are $\mathsf{negl}_5(\lambda)$-close for some negligible $\mathsf{negl}_5(\lambda)$. Note that the adversary $\mathcal{A}$ is not given Msk in G_4 and G_5, we conclude that

$$|\Pr[S_5] - \Pr[S_4]| \leq \mathsf{negl}_5(\lambda) \tag{7}$$

We construct an efficient algorithm $\mathcal{B}$ that solves the SIS problem. $\mathcal{B}$ receives a random matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ which is a $\mathsf{SIS}_{n,q,m,\beta}$ problem instance. $\mathcal{B}$ runs G_5 with the adversary $\mathcal{A}$ (the algorithm maintains all procedures and random oracles as per the specification of G_5). If G_5 outputs 0, $\mathcal{B}$ aborts. If G_5 outputs 1, i.e., the event S_5 happens, $\mathcal{A}$ must have outputted $(\mathsf{id}^*, \mathsf{m}^*, \sigma^*) \notin L_\sigma$ (i.e., σ^* is not produced by $\mathsf{OSig}(\mathsf{id}^*, \mathsf{m}^*)$) such that $\mathsf{Ver}(\mathsf{Mpk}, \mathsf{id}^*, \mathsf{m}^*, \sigma^*) = 1$ where $\mathsf{id}^* \notin L_{\mathsf{id}}$, assume $\sigma^* := (\sigma_1^*, \sigma_2^*) = (\mathbf{d}^* \in \mathbb{Z}^{m+w}, b^* \in \{0,1\})$ and $\mathbf{d}^* = \begin{bmatrix}\mathbf{d}_1^* \\ \mathbf{d}_2^*\end{bmatrix}$ with $\mathbf{d}_1^* \in \mathbb{Z}^m$ and $\mathbf{d}_2^* \in \mathbb{Z}^w$. $\mathcal{B}$ does:

1. Compute $\mu^* \leftarrow H'(\mathsf{id}^*)$, $\delta^* \leftarrow H''(\mathsf{id}^*||\mathsf{m}^*)$, and abort if $b^* \neq 1 - \mu^*$ or $\delta^* \neq 1 - \mu^*$; Otherwise
2. Return $\mathbf{e} = [\mathbf{I}_m|\mathbf{R}^*](\mathbf{d} - \mathbf{d}^*)$ where $\mathbf{R}^* = \mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id}^*||H'(\mathsf{id}^*)))$ as the SIS solution.

The following three lemmas demonstrate that with a probability close to $1/4$, $\mathbf{e}$ is a valid SIS solution, i.e., $\mathbf{A}\mathbf{e} = \mathbf{0}$, $\mathbf{e} \neq \mathbf{0}$ and $\|\mathbf{e}\| \leq \beta$.

Lemma 11. *The event $b^* = 1 - \mu^* = \delta^*$ happens with probability $1/4$.*

Proof. In G_5, the only oracles that may give $\mathcal{A}$ the information about $H'(\mathsf{id}^*)$ are H_1 and OExt. As shown in the proof of the inequality 3, the output distribution of H_1 is statistically close to $U(\mathbb{Z}_q^{n\times w})$. So, H_1 leaks no information about $H'(\mathsf{id}^*)$. Moreover, id^* is not allowed to query OExt as per the SEUF-ID-CMA definition. So, $\mu^* = H'(\mathsf{id}^*)$ remains random from $\mathcal{A}$'s view. Similarly, the only oracles may leak information about $\delta^* = H''(\mathsf{id}^*||\mathsf{m}^*)$ are H_2 and OSig. However, we have proved that the output distribution of H_2 is statistically close to $U(\mathbb{Z}_q^n)$, and bears no information about $H''(\mathsf{id}^*||\mathsf{m}^*)$. Also, $(\mathsf{id}^*, \mathsf{m}^*)$ is not allowed to query OSig as per the definition of SEUF-ID-CMA. Hence, δ^* remains random from $\mathcal{A}$'s view, and, $1 - b^* = \mu^* = 1 - \delta^*$ happens with probability $1/4$. □

Lemma 12. *Provided the condition $b^* = 1 - \mu^* = \delta^*$ holds, $\mathbf{A}\mathbf{e} = \mathbf{0}$.*

Proof. As the forged signature $\sigma^* = (\mathbf{d}^*, b^*)$ is valid, we must have

$$[\mathbf{A}|H_1(\mathsf{id}^*||b^*)]\mathbf{d}^* = H_2(\mathsf{id}^*||\mathsf{m}^*).$$

According to the simulation of H_2 in G_5 (i.e., Fig. 8),

$$[\mathbf{A}|H_1(\mathsf{id}^*||\delta^*)]\mathbf{d} = H_2(\mathsf{id}^*||\mathsf{m}^*)$$

where $\delta^* \leftarrow H''(\mathsf{id}^*||\mathsf{m}^*)$ and $\mathbf{d} = \mathsf{SampleR}(m+w, 1, s'; \tilde{H}(\mathsf{id}^*||\mathsf{m}^*))$. If $b^* = 1-\mu^* = \delta^*$, according to the simulation of H_1 in G_5 (i.e., Fig. 7), $H_1(\mathsf{id}^*||b^*) = H_1(\mathsf{id}^*||\delta^*) = \mathbf{A}\mathbf{R}^*$ where $\mathbf{R}^* = \mathsf{SampleR}(m, w, s; \hat{H}(\mathsf{id}^*||1-\mu^*))$. Consequently, we obtain $[\mathbf{A}|\mathbf{A}\mathbf{R}^*](\mathbf{d}-\mathbf{d}^*) = \mathbf{0}$ and thus $\mathbf{A}\mathbf{e} = \mathbf{0}$. □

Lemma 13. *Let* $\mathbf{e}$ *be constructed as above. We have* $\mathbf{e} \neq \mathbf{0}$ *and* $\|\mathbf{e}\| \leq \beta$ *where* $\beta = O(w^{3/2}) \cdot \omega(\sqrt{\log n})^2$ *with all but negligible probability.*

Proof. To show that $\|\mathbf{e}\| \leq \beta$, we have

$$\begin{aligned}\|\mathbf{e}\| &\leq \|(\mathbf{d}_1 - \mathbf{d}_1^*)\| + s_1(\mathbf{R}_{\mathsf{id}^*||b^*}) \cdot \|\mathbf{d}_2 - \mathbf{d}_2^*\| \\ &\leq 2s'\sqrt{m} + (C \cdot s(\sqrt{m}+\sqrt{w})) \cdot 2s'\sqrt{w} \\ &\leq O(w^{3/2}) \cdot \omega(\sqrt{\log n})^2 = \beta\end{aligned}$$

To show $\mathbf{e} \neq \mathbf{0}$ w.h.p. Let $\mathbf{d} = \begin{bmatrix}\mathbf{d}_1 \\ \mathbf{d}_2\end{bmatrix}$ where $\mathbf{d}_1 \in \mathbb{Z}^m$, $\mathbf{b}_2 \in \mathbb{Z}^w$. We consider the following cases:

- $\mathbf{x} = \mathbf{d}_2 - \mathbf{d}_2^* = \mathbf{0}$: In this case, $\mathbf{y} = \mathbf{d}_1 - \mathbf{d}_1^* \neq \mathbf{0}$. If $\mathcal{A}$ has queried $(\mathsf{id}^*, \mathsf{m}^*)$ to OSig and received $(\mathbf{d}, \delta^*) = (\mathbf{d}, b^*)$ as the answer (note this is consistent with the simulation of OSig), we must have $\mathbf{d} \neq \mathbf{d}^*$ and thus, $\mathbf{e} \neq \mathbf{0}$.
- $\mathbf{y} = \mathbf{d}_1 - \mathbf{d}_1^* \neq \mathbf{0}$: In this case, $\mathbf{y}$ must have a non-zero coordinate. W.l.o.g, assume last coordinate of $\mathbf{y}$, y is non-zero. Then, $\mathbf{e}$ can be written as $\mathbf{e} = \mathbf{x} + \mathbf{r}^* y + \mathbf{v}$ where $\mathbf{r}^*$ is the last column of $\mathbf{R}^*$, $\mathbf{v} \in \mathbb{Z}^m$ depends on $\mathbf{R}^*$'s remaining columns and $\mathbf{y}$'s remaining coordinates. Note, $\mathbf{R}^*$ (also $\mathbf{r}^*$) is not given to the adversary $\mathcal{A}$. The only information about $\mathbf{r}^*$ that $\mathcal{A}$ can get is via $H_1(\mathsf{id}^*||b^*) = \mathbf{A}\mathbf{R}^*$. By Lemma 10, $H_1(\mathsf{id}^*||b^*)$ is statistically close to $U(\mathbb{Z}_q^{n\times w})$ w.h.p. So, conditioned on $H_1(\mathsf{id}^*||b^*) = \mathbf{A}\mathbf{R}^*$, $\mathbf{r}^*$ is distributed negligibly close to $D_{\mathbb{Z}^m, s}$. By Lemma 5, $\mathbf{r}^*$ has min-entropy $\geq m-1$. So, $\mathbf{e} \neq \mathbf{0}$ with probability $1 - 2^{-(m-1)} = 1 - \mathsf{negl}_6'(\lambda)$ for some negligible $\mathsf{negl}_6'(\lambda)$. □

Let E be the event that the algorithm $\mathcal{B}$ solves the SIS problem challenge. Combining Lemma 12, Lemma 13 and Lemma 11 leads to

$$\begin{aligned}\Pr[E] &= \Pr[E|S_5]\Pr[S_5] + \Pr[E|\neg S_5]\Pr[\neg S_5] \\ &\geq \Pr[E|S_5]\Pr[S_5] = \frac{1}{4}(1 - \mathsf{negl}_6'(\lambda))\Pr[S_5] \\ &= \frac{1}{4}\Pr[S_5] - \mathsf{negl}_6'(\lambda) \cdot \Pr[S_5]\end{aligned}$$

Note that $\Pr[E] \leq \mathsf{Adv}^{\mathsf{SIS}_{n,q,m,\beta}}_{\mathcal{A}}(\lambda)$. Thus,

$$\Pr[S_5] \leq 4 \cdot \mathsf{Adv}^{\mathsf{SIS}_{n,q,m,\beta}}_{\mathcal{B}}(\lambda) + \mathsf{negl}_6(\lambda) \tag{8}$$

for negligible $\mathsf{negl}_6(\lambda) = 4 \cdot \mathsf{negl}'_6(\lambda) \cdot \Pr[S_5]$. Putting the inequalities 2, 3, 4, 5, 6, 7, and 8 together, we have

$$\mathsf{Adv}^{\mathsf{seuf}}_{\Sigma',\mathcal{A}}(\lambda) \leq 4 \cdot \mathsf{Adv}^{\mathsf{SIS}_{n,q,m,\beta}}_{\mathcal{B}}(\lambda) + \mathsf{negl}'(\lambda)$$

where $\mathsf{negl}'(\lambda)$ upper bounds the negligible terms $\sum_{i=1}^{6} \mathsf{negl}_i(\lambda)$. This concludes the SEUF-ID-CMA security proof for the scheme Σ'. □

4.3 The Security of IBS Scheme Σ

Proof. Combining the proofs from Section 4.1 and Section 4.2 results in

$$\begin{aligned}\mathsf{Adv}^{\mathsf{seuf}}_{\Sigma,\mathcal{A}}(\lambda) &\leq 2(Q_{\mathsf{Ext}} + Q_{\mathsf{Sig}}) \cdot 2^{-\ell/2} + \mathsf{Adv}^{\mathsf{seuf}}_{\Sigma',\mathcal{A}}(\lambda) \\ &\leq 4 \cdot \mathsf{Adv}^{\mathsf{SIS}_{n,q,m,\beta}}_{\mathcal{B}}(\lambda) + \mathsf{negl}(\lambda)\end{aligned}$$

where the negligible $\mathsf{negl}(\lambda)$ upper bounds $\mathsf{negl}'(\lambda) + 2(Q_{\mathsf{Ext}} + Q_{\mathsf{Sig}}) \cdot 2^{-\ell/2}$. Let the running time of the attacker $\mathcal{A}$ and the time used by simulations in the proofs be t' and t. t is bounded by $t' + (Q_{H_1} + Q_{H_2}) \cdot \mathsf{poly}(\lambda)$ as proof simulations run $\mathcal{A}$ as a subroutine and the extra computations of the simulation can be done in polynomial time. □

5 Security Proof (Sketch) in the Random Oracle Model

Theorem 2. *Let H_1, H_2, H_1, H_2, H_3 be random oracles. If (t, ϵ)-$\mathsf{SIS}_{n,q,m,\beta}$ assumption holds with $\beta = O(w^{3/2})\omega(\sqrt{\log n})^2$, the IBS scheme Σ is $(t', Q_{\mathsf{Ext}}, Q_{\mathsf{Sig}}, 4\epsilon + \mathsf{negl}(\lambda))$-SEUF-ID-CMA secure for $t' \leq t + O(Q_{H_1} + Q_{H_2}) \cdot \mathsf{poly}(\lambda)$ where Q_{H_1} and Q_{H_2} are the numbers of random oracle queries to $|H_1\rangle$ and $|H_2\rangle$.*

Proof (Sketch). The theorem states the security of our IBS scheme in the classic ROM. We sketch the security proof due to the similarity to the QROM proof. Similar to the QROM proof, we first reduce the SEUF-ID-CMA security of the IBS scheme Σ (Fig. 2) to the stateful IBS scheme Σ' defined in Fig. 3. This step replaces the random oracles H_1, H_2, H_3 by internal states, and uses Lemma 2 to get inequality 1, showing the security gap between Σ and Σ' is negligible.

Then, we construct a tight reduction $\mathcal{B}$ in Fig 11 that converts the adversary $\mathcal{A}$'s advantage agaisnt the stateful IBS scheme Σ' to its advantage in solving the SIS problem. On receiving the SIS challenge $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, $\mathcal{B}$ runs the sub-routines $\mathsf{Setup}_{\mathcal{B}}$, $\mathsf{Sig}_{\mathcal{B}}$, $\mathsf{Ext}_{\mathcal{B}}$, and maintains the random oracles H_1, H_2 to simulate the IBS scheme Σ' perfectly. Finally, it uses the sub-routine $\mathsf{Sol}_{\mathcal{B}}$ to extract the SIS problem solution $\mathbf{e}$. Like the proof in the QROM, the factor 4 in the security loss comes from requiring the condition $\mu^* = 1 - b^* = 1 - \delta^*$ holds.

Finally, similar to the QROM proof, the running time of the first-step reduction is nearly the same as the corresponding adversary, and the running time of the reduction $\mathcal{B}$'s running time t is bounded by $t' + (Q_{H_1} + Q_{H_2}) \cdot \mathsf{poly}(\lambda)$ where t' is the running time against the statful IBS scheme Σ'. □

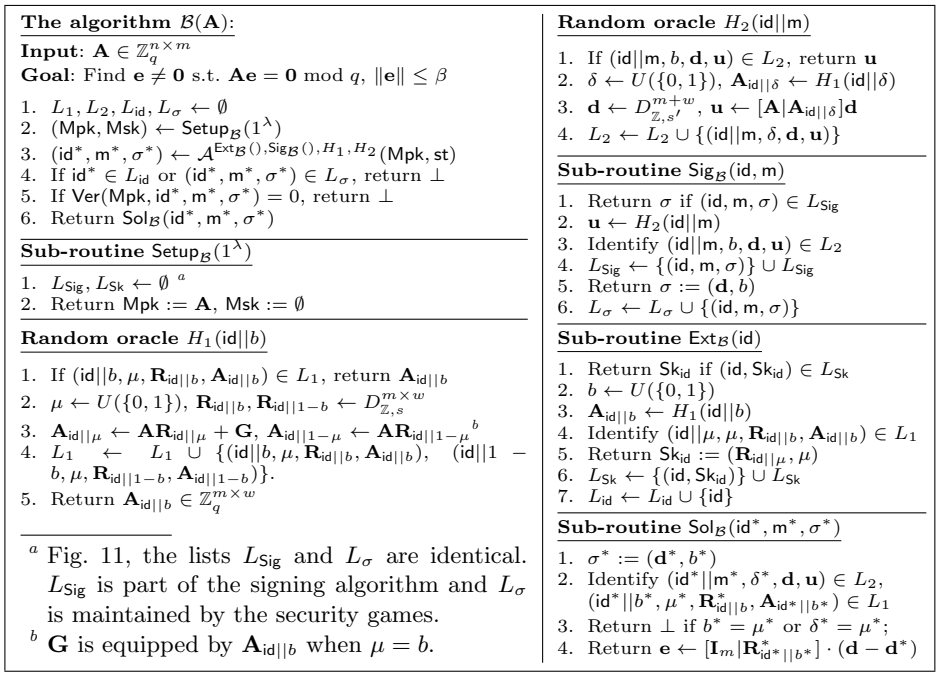


Fig. 11: Reduction to SIS in Random Oracle Model

6 Conclusion

We have presented a short lattice IBS scheme. We proved, in detail, that the scheme is strongly existentially unforgeable under the chosen-identity attack and the chosen-message attack based on the standard SIS problem with a tight security reduction in the QROM and ROM. Our scheme outperforms the existing schemes with shorter signatures and a stronger unforgeability and relying on weaker SIS assumptions. We leave constructing a short and tightly quantum-secure IBS scheme in the plain model as a future research question.

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